Requirements specification for Kindergarten Management System business process

1. General description of business process

a. A general description of the business process and a description of the performance metrics generated by this process, possible current analytical problems.

The kindergarten meal planning process begins with estimating the required food quantities based on the number of enrolled children. The system calculates the necessary ingredients by considering planned meals and portion sizes. Each meal consists of specific ingredients, for example, 30g of oats, 60ml of milk, and half a banana for breakfast.

Once the required ingredients are determined, the kindergarten orders food supplies in bulk, categorized by product type (e.g., grains, dairy, fruits). The kitchen staff receives and stores the ingredients, ensuring proper storage conditions.

On the day of meal preparation, the system picks a meal plan, detailing portion sizes and required ingredient quantities. The staff prepares meals accordingly and records the total number of prepared meals. During mealtime, each child receives a meal, and the staff notes the number of meals distributed. Some meals might not be taken due to absences or refusals. At the end of the meal session, the staff records:

- The number of meals left uneaten (waste).
- The number of meals given to children (consumed).
- The number of meals prepared (initial count).

This data is analysed to adjust future meal orders and minimize food waste. If excessive waste is observed, the kindergarten may adjust portion sizes, change meal preferences, or reduce ingredient purchases accordingly.

The food waste percentage decreases by 5% compared to the previous year.

The total food expenditure decreases by 5% through optimized meal planning and portion control.

b. Typical questions

1. Compare the amount of food waste between different meal types (e.g., breakfast vs. lunch vs. snacks).
2. Compare the number of uneaten meals between different kindergarten groups.
3. Compare the amount of ingredients used this month with the previous month.
4. Were there any days this month when more than 10 meals were left uneaten?
5. Did any ingredients (if in meal) have smaller influence on waste level?
6. What is the average number of uneaten meals per day over the last month?
7. What percentage of prepared meals were fully consumed in the last quarter?

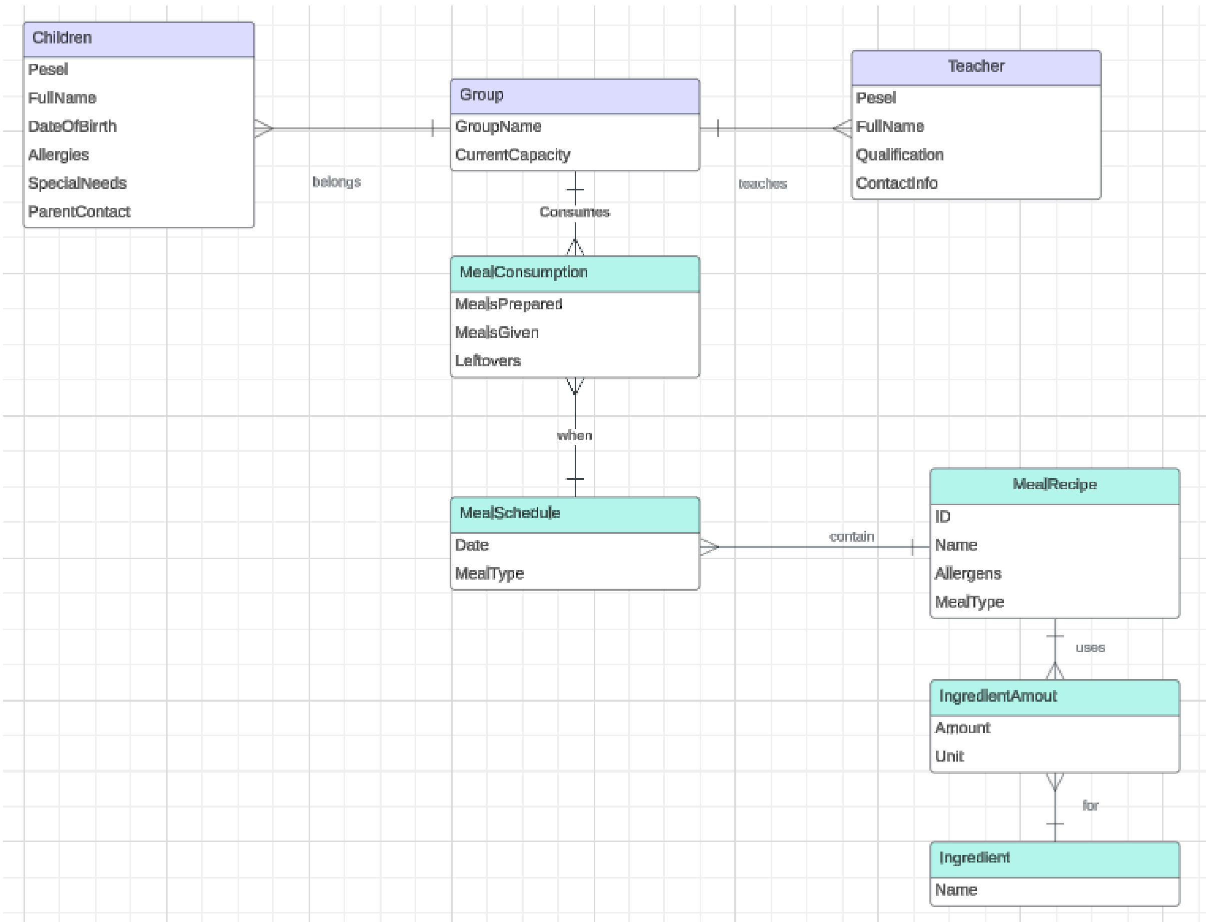
8. How has the amount of food waste changed over the past six months?
9. Is there a seasonal trend in the consumption of certain meals (e.g., do children eat more fruit in summer)?
10. Rank the most wasted meal in the last three months.

c. Data

Data is stored in a kindergarten management system - meal consumption is tracked there (meals prepared, meals given, leftovers) taking into account different age groups as well as meal schedule (date, meal type). Meal recipes are also stored there (name, allergens, meal type) with addition in Ingredient (name) and IngredientsAmount (amount, unit). Information about ordering ingredients is stored in a CSV file including date, type, name, amount, price per unit and price for the whole product.

2. Data sources structures

Kindergarten Management System



Children (Pesel, FullName, DateOfBirth, Allergies, SpecialNeeds, ParentContact, GroupName REF Group)

Teacher (Pesel, FullName, Qualification, ContactInfo, GroupName REF Group)

Group (GroupName, CurrentCapacity)

MealConsumption(MealsPrepared, MealsGiven, Leftovers, GroupName REF Group, (Date, MealType) REF MealSchedule)

MealSchedule(Date, MealType, ID REF MealRecipe)

MealRecipe (ID, Name, Allergens, MealType)

IngredientAmount (Amount, Unit, ID REF MealRecipe, Name REF Ingredient)

Ingredient (Name)

Table name	Attribute	Attribute type	Description
Children	**Information about each child attending kindergarten**		
	Pesel	String – 11 characters	PK, (BK)
	FullName	String	Name and surname
	DateOfBirth	Date	date of birth
	Allergies	String	Allergies of child
	SpecialNeeds	String	Any meds needed
	ParentContact	String - 9 digits	Phone number to the parent
Group	**Information about groups to which children are divided**		
	GroupName	String	PK, (BK)
	CurrentCapacity	INT	Current number of children in group
Teacher	**Information about teachers working in kindergarten**		
	Pesel	String – 11 characters	PK, (BK)
	FullName	String	Name and surname
	Qualification	String	Educational

			background, certifications or professional expertise
	ContactInfo	String - 9 digits	Phone number
MealConsumption	**Information about meals consumed by groups**		
	MealsPrepared	INT	How many meals have been prepared
	MealsGiven	INT	How many meals have been given to children
	Leftovers	INT	How many meals have been left uneaten
	GroupName	String	FK, part of PK (BK)
	Date, MealType	-	FK, part of PK (BK)
MealSchedule	**What meals were prepared on that day**		
	Date	Date	When event occurred, part of PK (BK)
	MealType	String	Type of meal (breakfast, lunch …), part of PK (BK)
MealRecipe	**Meal recipes**		
	ID	INT	PK
	Name	String	Name of meal, (BK)
	Allergens	String	List of allergens in meal
	MealType	String	Type of meal, possible values: breakfast, snack, lunch, dinner
Ingredient	**Information about ingredients used in recipes**		
	Name	String	PK, Name of ingredient, (BK)
IngredientAmount	**Information about amounts of ingredients used in recipes**		
	Amount	INT	Amount of

			ingredient
	Unit	String	Unit of used ingredient, like grams, millilitres
	ID	INT	FK, part of PK, (BK)
	Name	String	FK, part of PK, (BK)

Note: New children can be assigned to the group during a month, every year children move from one group to another (3 age group → 4 age group, 4 → 5). Teachers do not rotate, they are always assigned to a particular group, the only special case is firing a teacher.

Order list CSV

Order list (Information about ordering ingredients, each line describes one ingredient, line 1 is a header row):

Column 1: Date (Date in format yyyy-mm-dd),
Column 2: Type (String, [‘fruit’, ‘vegetable’, ‘dairy’, ‘meat’, ‘sides/other’]),
Column 3: Name (String, like “Milk”),
Column 4: Amount (Numeric, 1 decimal precision),
Column 5: Price per unit (Numeric, 2 decimal precision)

Note: Each line does not mean separate order. It can be, for example, 5 ingredients delivered in one go.

3. Scenarios of analytical problems

Why was there an increase or decrease in meal waste across 3 months?

1. Compare total meals prepared vs. meals given vs. leftovers for each meal type in the past three months.
2. How does the total cost of ingredients in a single portion affect the meal waste?
3. Compare waste levels for meals with different portion sizes across 3 months.
4. Identify meal types with the highest increase/decrease in waste over the last three months.
5. Compare total cost of ingredients vs. the number of meals prepared.
6. Does weather (temperature, seasonal changes) correlate with changes in meal waste?
7. Survey-based feedback: Did children enjoy specific meals, and does enjoyment correlate with waste levels?

What is the effect of meal portion and ingredient adjustments on food waste reduction

1. Does having more/fewer products of a certain type affect the waste reduction?
2. What is the most common ingredient that can be found in a meal with the highest waste level?
3. Calculate the cost of meals wasted due to leftovers from the last 3 months and calculate rate of change.
4. Identify if removing or substituting ingredients affected waste levels.
5. Compare waste levels for meal types that were adjusted vs. those that were not.
6. Check if nutrition guidelines (external standards) align with portion sizes used.
7. Did teachers' supervision or mealtime encouragement impact meal consumption?

4. Data needed for analytical problems

Analytical problem: "*Why was there an increase or decrease in meal waste across 3 months?"*

1. Compare total meals prepared vs. meals given vs. leftovers for each meal type in the past three months.
   - **Meals Prepared, Meals Given, Leftovers**: DB - table `MealConsumption`, columns `MealsPrepared`, `MealsGiven`, `Leftovers`
   - **Meal Type**: DB - table `MealSchedule`, column `MealType`
   - **Date**: DB - table `MealSchedule`, column `Date`
2. How does the total cost of ingredients in a single portion affect the meal waste?
   - **Leftovers**: DB - table `MealConsumption`, columns `Leftovers`
   - **Ingredients, Amounts**: DB - table `IngredientAmount and Ingredient`, column `Name`, `Amount`,
   - **Cost per unit**: CSV - columns `Name`, `Price per unit`
3. Compare waste levels for meals with different portion sizes across 3 months.
   - **Leftovers**: DB - table `MealConsumption`, column `Leftovers`
   - **Portion Sizes**: DB - table `IngredientAmount and Ingredient`, columns `Name`, `Amount`
   - **Date**: DB - table `MealSchedule`, column `Date`
4. Identify meal types with the highest increase/decrease in waste over the last three months.
   - **Leftovers**: DB - table `MealConsumption`, column `Leftovers`
   - **Meal Type**: DB - table `MealSchedule`, column `MealType`
   - **Date**: DB - table `MealSchedule`, column `Date`
5. Compare total cost of ingredients vs. the number of meals prepared.
   - **Meals Prepared**: DB - table `MealConsumption`, column `MealsPrepared`
   - **Ingredient Costs**: CSV - columns `Date`, `Name`, `Amount`, `Price per unit`
6. Does weather (temperature, seasonal changes) correlate with changes in meal waste?
   - **Leftovers**: DB - table `MealConsumption`, column `Leftovers`
   - **Weather Data**: External weather data source, including `Date`, `Temperature`, and `Season`

7. Survey-based feedback: Did children enjoy specific meals, and does enjoyment correlate with waste levels?
    - **Leftovers**: DB - table `MealConsumption`, column `Leftovers`
    - **Meal Feedback**: Newly introduced survey system collecting meal preference feedback, columns such as `MealName`, `Enjoyment` (scale 1-5)

Analytical problem: *"What is the effect of meal portion and ingredient adjustments on food waste reduction?"*

1. Does having more/fewer products of a certain type affect the waste reduction?
    - **Leftovers**: DB - table `MealConsumption`, column `Leftovers`
    - **Product type**: CSV - column `Type`
2. What is the most common ingredient that can be found in a meal with the highest waste level?
    - **Leftovers, Meals Prepared**: DB - table `MealConsumption`, column `Leftovers, MealsPrepared`
    - **Ingredient**: DB - table `Ingredient`, column Name
3. Calculate the cost of meals wasted due to leftovers from the last 3 months and calculate rate of change.
    - **Ingredient Costs**: CSV - columns Name, `Price per unit`
    - **Leftovers**: DB - table `MealConsumption`, column `Leftovers`
    - **`Ingredients`**`: DB - table IngredientAmount and Ingredient, column Amount, Name`
4. Identify if removing or substituting ingredients affected waste levels.
    - **Leftovers**: DB - table `MealConsumption`, column `Leftovers`
    - **Ingredient Substitutions**: DB - table `IngredientAmount` and `Ingredient`, columns Name, `Amount` (changed recipies)
    - **Date**: DB - table `MealSchedule`, column `Date`
5. Compare waste levels for meal types that were adjusted vs. those that were not.
    - **Leftovers**: DB - table `MealConsumption`, column `Leftovers`
    - **Meal Types**: DB - table `MealSchedule`, column `MealType`
6. Check if nutrition guidelines (external standards) align with portion sizes used.
    - **Portion Sizes**: DB - table `IngredientAmount`, columns Name, `Amount`
    - **Nutrition Standards**: External nutritional guidelines data source
7. Did teachers' supervision or mealtime encouragement impact meal consumption?
    - **Meals Given**: DB - table `MealConsumption`, column `MealsGiven`
    - **Teacher Interactions**: Newly introduced tracking system for teacher encouragement during meals